

atlántida

Formación Profesional

Práctica Unidad 2. Introducción a los servidores web

ÍNDICE

-Instalar Ubuntu Desktop

-Instalación paquete LAMP Ubuntu

-Instalando NGINX

-Instalación Ubuntu Server

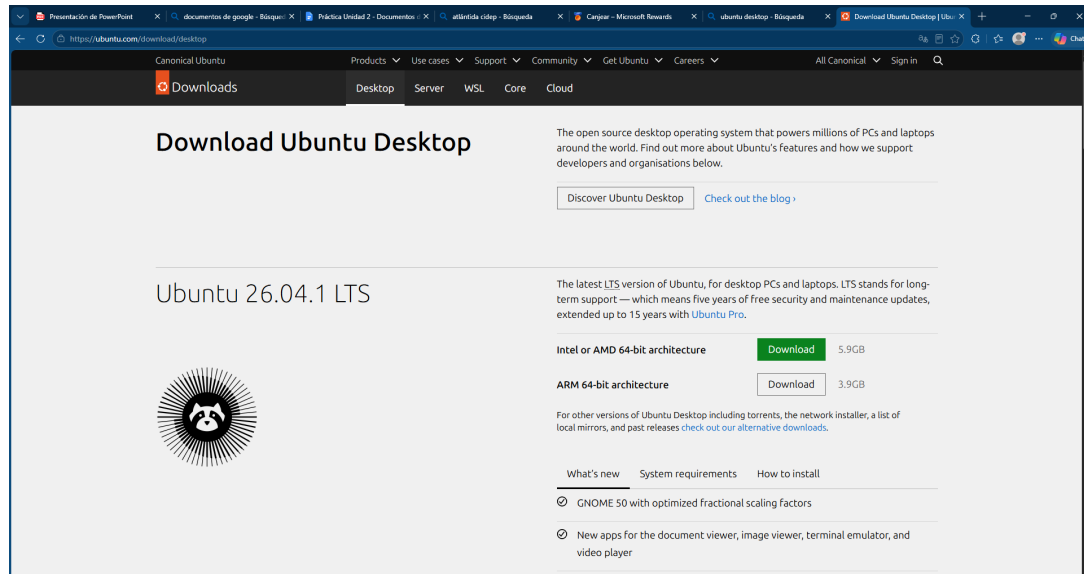
-Instalacion Windows Server

-Instalacion IIS en Windows

- Instala Docker en Ubuntu y despliega un servidor web Apache que escuche por el puerto 8081.

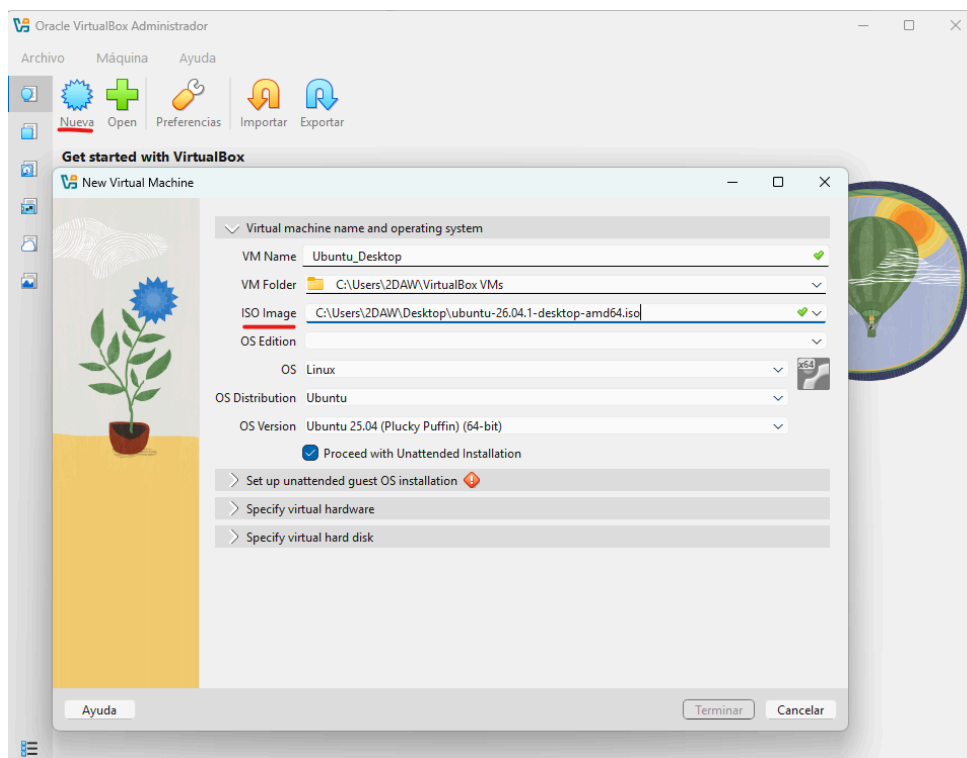
-INSTALACION UBUNTU DESKTOP

Instalación de Ubuntu Desktop en la página oficial

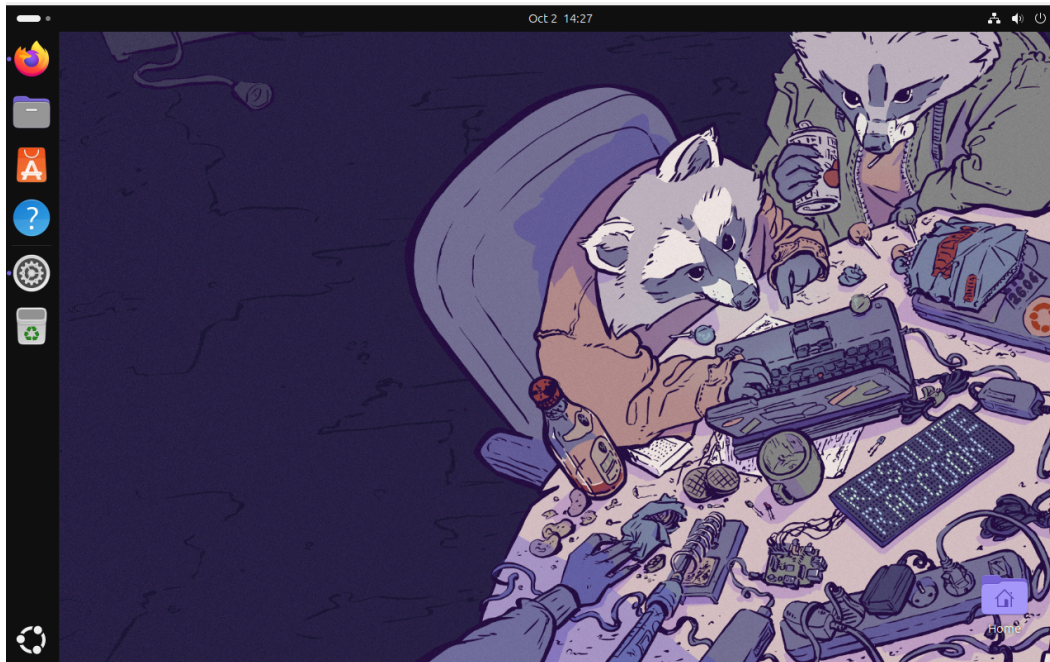


 ubuntu-26.04.1-desktop-amd64	29/09/2026 16:01	Archivo de image...
 ubuntu-26.04.1-live-server-amd64	29/09/2026 16:05	Archivo de image...

En virtualBox nos dirigimos a “Nueva” y agregamos la ISO Image de Ubuntu Desktop.



Iniciamos la máquina virtual, y acto seguido hacemos skip, y ya estaremos en el escritorio de nuestra máquina virtual.



-INSTALACIÓN PAQUETE LAMP UBUNTU

1. Actualizar el sistema

```
vboxuser@ubuntuDesktop:~  
vboxuser@UbuntuDesktop:~/Videos$ cd  
vboxuser@UbuntuDesktop:~$ sudo apt update 66  
[sudo: authenticate] Password:  
sudo: Authentication failed, try again.  
[sudo: authenticate] Password:  
sudo: Authentication failed, try again.  
[sudo: authenticate] Password:  
sudo: maximum 3 incorrect authentication attempts  
vboxuser@UbuntuDesktop:~$ sudo apt update && sudo apt upgrade -y
```

2. Instalar Apache

```
vboxuser@UbuntuDesktop:~$ sudo apt install apache2 -y
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-30          linux-modules-7.0.0-30-generic
  linux-headers-7.0.0-30-generic  linux-tools-7.0.0-30
  linux-image-unsigned-7.0.0-30-generic  linux-tools-7.0.0-30-generic
  linux-main-modules-zfs-7.0.0-30-generic
Use 'sudo apt autoremove' to remove them.

Installing:
  apache2

Installing dependencies:
  apache2-bin  apache2-utils  libaprutil1-dbd-sqlite3  libaprutil1t64
  apache2-data  libapr1t64      libaprutil1-ldap

Suggested packages:
  apache2-doc  apache2-suexec-pristine | apache2-suexec-custom

Summary:
```

3. Instalar MySQL

```
Processing triggers for libc-bin (2.43-2ubuntu2.4)...
vboxuser@UbuntuDesktop:~$ sudo apt install mysql-server -y
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-30          linux-modules-7.0.0-30-generic
  linux-headers-7.0.0-30-generic  linux-tools-7.0.0-30
  linux-image-unsigned-7.0.0-30-generic  linux-tools-7.0.0-30-generic
  linux-main-modules-zfs-7.0.0-30-generic
Use 'sudo apt autoremove' to remove them.

Installing:
  mysql-server

Installing dependencies:
  libaio1t64      libfcgi0t64      libtcmalloc-minimal4t64
  libcgi-fast-perl  libgoogle-perftools4t64  mysql-client
  libcgi-pm-perl   libhtml-template-perl  mysql-client-core
  libfcgi-bin      libmecab2         mysql-common
```

Ejecuta el script de seguridad para configurar contraseñas y accesos:

```
vboxuser@UbuntuDesktop:~$ sudo mysql_secure_installation
[sudo: authenticate] Password:

Securing the MySQL server deployment.

Connecting to MySQL using a blank password.

VALIDATE PASSWORD COMPONENT can be used to test passwords
and improve security. It checks the strength of password
and allows the users to set only those passwords which are
secure enough. Would you like to setup VALIDATE PASSWORD component?

Press y|Y for Yes, any other key for No: y
```

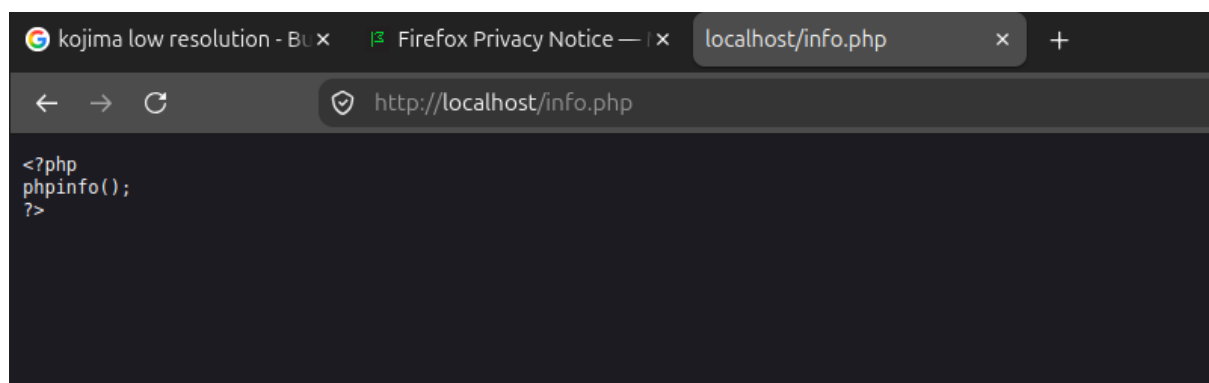
Instalar PHP y módulos comunes

```
vboxuser@UbuntuDesktop:~$ sudo apt install php lbapache2-mod-php php-mysql -y
```

5. Reiniciar Apache

```
vboxuser@UbuntuDesktop:~$ sudo systemctl restart apache2
vboxuser@UbuntuDesktop:~$
```

Verificar la instalación de PHP



-INSTALACIÓN NGINX

Interested in NGINX's open source Kubernetes ecosystem? Start here: kubernetes.nginx.org

nginx: download

Mainline version

CHANGES	nginx-1.31.6	nginx/Windows-1.31.6	pgp
	pgp		

Stable version

CHANGES	nginx-1.30.5	nginx/Windows-1.30.5	pgp
1.30	pgp		

Legacy versions

CHANGES	nginx-1.28.3	nginx/Windows-1.28.3	pgp
1.28	pgp		
CHANGES	nginx-1.26.3	nginx/Windows-1.26.3	pgp
1.26	pgp		
CHANGES	nginx-1.24.0	nginx/Windows-1.24.0	pgp
1.24	pgp		
CHANGES	nginx-1.22.1	nginx/Windows-1.22.1	pgp
1.22	pgp		
CHANGES	nginx-1.20.2	nginx/Windows-1.20.2	pgp
1.20	pgp		



english
[русский](#)

[news](#)
[about](#)
[download](#)
[security](#)
[documentation](#)
[faq](#)
[books](#)
[community](#)
[enterprise](#)

[community forum \(new\)](#)
[x.com](#)
[blog](#)

[njs](#)
[ingress controller](#)
[gateway fabric](#)

Antes de proceder con la instalación debemos de comprobar que todos los archivos están actualizados a su última versión.

```
ubuntu@UbuntuServer:~$ sudo apt update && sudo apt upgrade -y
```

Una vez comprobado que todo esta actualizado procedemos a instalarlo con el siguiente comando

```
ubuntu@UbuntuServer:~$ sudo apt install nginx -y
```

Ahora comprobamos el estado en el que se encuentra el servicio.

```
ubuntu@UbuntuServer:~$ sudo systemctl status nginx
• nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-09-30 14:54:09 UTC; 2min 0s ago
     Invocation: c732fbbec294e8eb8572b679219308a
       Docs: man:nginx(8)
    Process: 7845 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
    Process: 7847 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Main PID: 7880 (nginx)
      Tasks: 3 (limit: 1730)
     Memory: 2.9M (peak: 6.9M)
        CPU: 33ms
   CGroup: /system.slice/nginx.service
           └─7880 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             └─7882 "nginx: worker process"
               └─7883 "nginx: worker process"

Sep 30 14:54:09 UbuntuServer systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Sep 30 14:54:09 UbuntuServer systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
ubuntu@UbuntuServer:~$
```

-INSTALACIÓN UBUNTU SERVER

Ubuntu canónico

Productos Casos de uso Apoyo Comunidad Consigue Ubuntu Carreras Todos canónicos Iniciar sesión

Descargas Escritorio **Servidor** WSL Núcleo Cloud

Consigue Ubuntu Server

Instalación manual Máquinas virtuales instantáneas Provisión automatizada

Ubuntu 26.04.1 LTS



La última versión LTS de Ubuntu Server. LTS significa soporte a largo plazo — lo que significa cinco años de actualizaciones gratuitas de seguridad y mantenimiento, extendidas hasta 15 años con [Ubuntu Pro](#).

Arquitectura Intel o AMD de 64 bits [Descargar](#) 3 GB

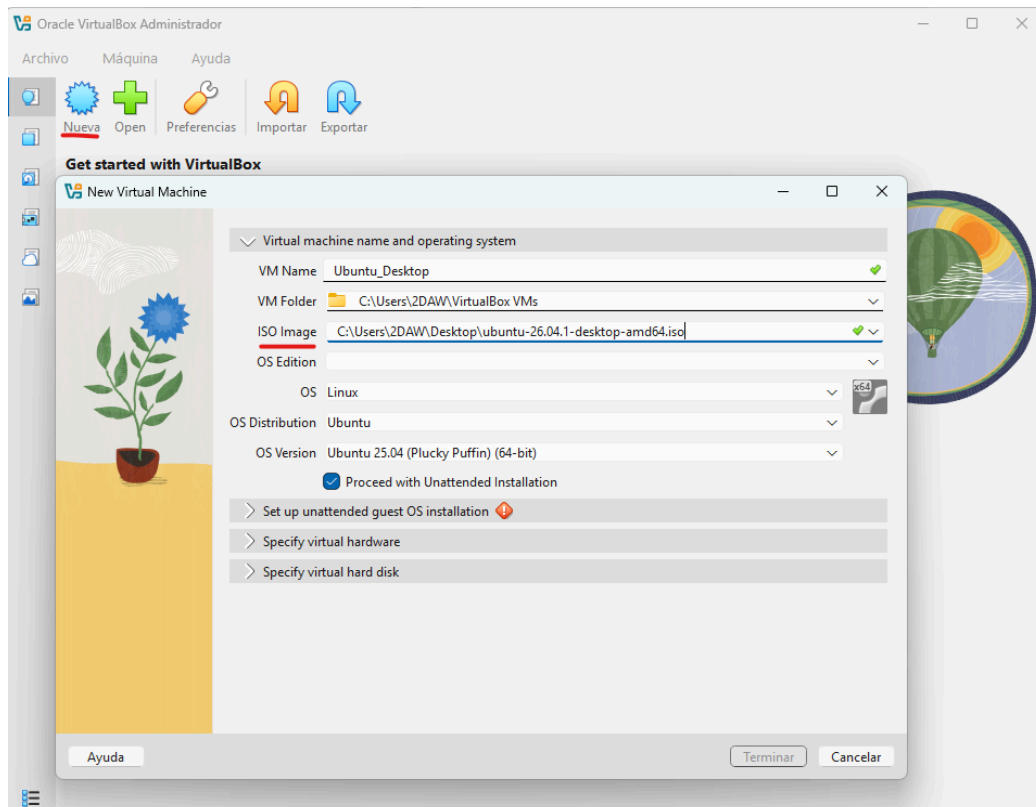
Para otras versiones de Ubuntu Server, incluyendo torrents, el instalador de red, una lista de espejos locales y versiones anteriores, [consulta nuestras descargas alternativas](#).
[Encuentra nuestras arquitecturas alternativas aquí](#).

Qué hay de nuevo Requisitos del sistema Cómo instalar

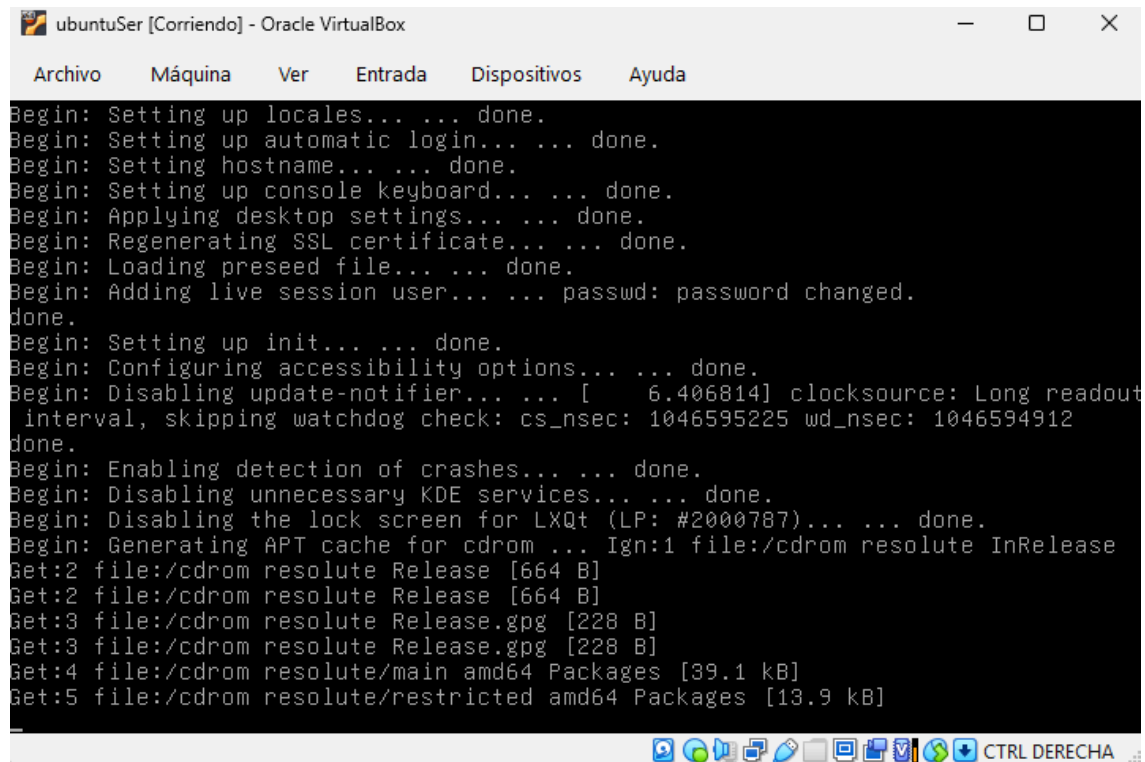
- ☑ Núcleo de Linux 7.0
- ☑ Sistema de doble pista para pilas de contenedores: tradicional con actualizaciones mayores regulares, o estable para mantener versiones consistentes de los componentes principales
- ☑ Valkey 9, PostgreSQL 18, MySQL y MariaDB como versiones soportadas a largo plazo, y la introducción de DocumentDB

 ubuntu-26.04.1-desktop-amd64	29/09/2026 16:01	Archivo de image...
 ubuntu-26.04.1-live-server-amd64	29/09/2026 16:05	Archivo de image...

Realizamos el mismo paso agregando una maquina virtual, esta vez con la iso de la versión Server.



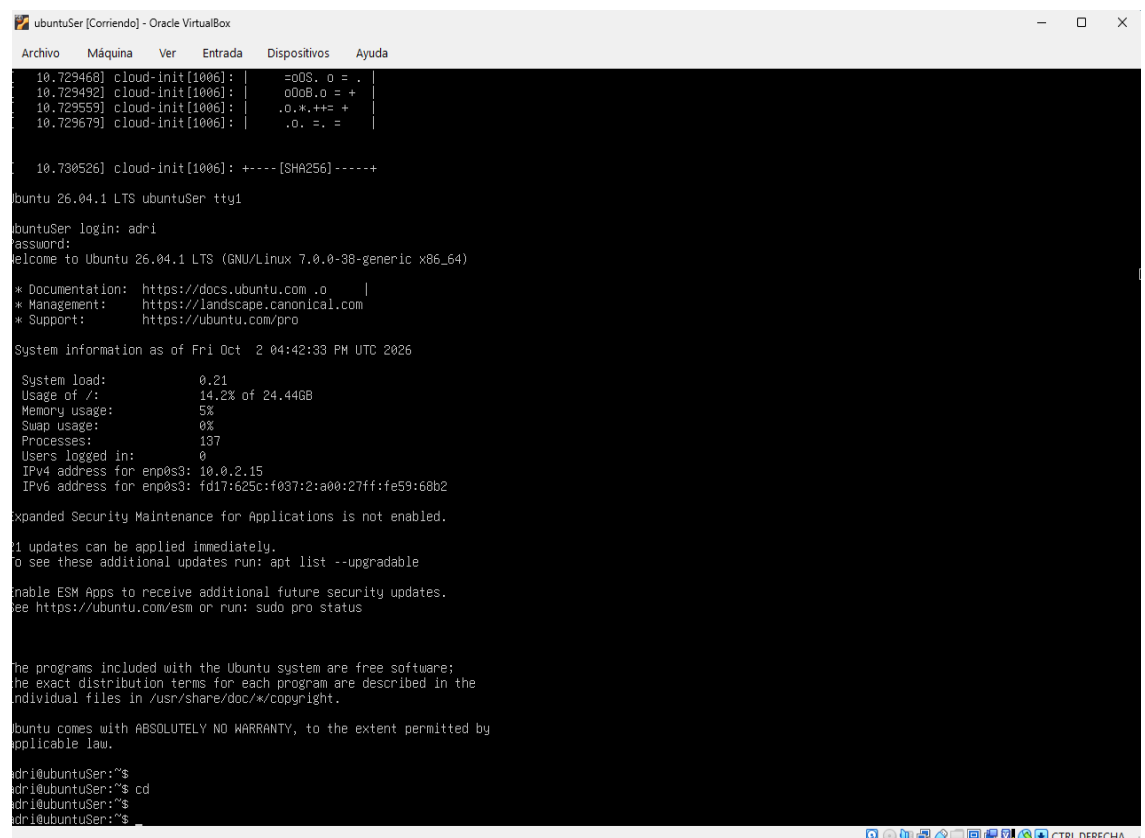
Proceso de instalación....



The screenshot shows the 'ubuntuSer [Corriendo] - Oracle VirtualBox' window. The terminal displays the progress of the Ubuntu Server installation. The menu bar includes 'Archivo', 'Máquina', 'Ver', 'Entrada', 'Dispositivos', and 'Ayuda'. The terminal output shows various system configuration steps being completed, such as setting up locales, automatic login, hostname, console keyboard, desktop settings, SSL certificate, preseed file, live session user, and init. It also shows the disabling of update-notifier and the enabling of crash detection. The installation is progressing through the 'resolute' release, downloading files from the CDROM.

```
Begin: Setting up locales... .. done.
Begin: Setting up automatic login... .. done.
Begin: Setting hostname... .. done.
Begin: Setting up console keyboard... .. done.
Begin: Applying desktop settings... .. done.
Begin: Regenerating SSL certificate... .. done.
Begin: Loading preseed file... .. done.
Begin: Adding live session user... .. passwd: password changed.
done.
Begin: Setting up init... .. done.
Begin: Configuring accessibility options... .. done.
Begin: Disabling update-notifier... .. [ 6.406814] clocksource: Long readout
interval, skipping watchdog check: cs_nsec: 1046595225 wd_nsec: 1046594912
done.
Begin: Enabling detection of crashes... .. done.
Begin: Disabling unnecessary KDE services... .. done.
Begin: Disabling the lock screen for LXQt (LP: #2000787)... .. done.
Begin: Generating APT cache for cdrom ... Ign:1 file:/cdrom resolute InRelease
Get:2 file:/cdrom resolute Release [664 B]
Get:2 file:/cdrom resolute Release [664 B]
Get:3 file:/cdrom resolute Release.gpg [228 B]
Get:3 file:/cdrom resolute Release.gpg [228 B]
Get:4 file:/cdrom resolute/main amd64 Packages [39.1 kB]
Get:5 file:/cdrom resolute/restricted amd64 Packages [13.9 kB]
```

Y listo los logueamos y ya tenemos instalado Ubuntu Server.



The screenshot shows the 'ubuntuSer [Corriendo] - Oracle VirtualBox' window. The terminal displays the login process for 'adri' on 'Ubuntu 26.04.1 LTS ubuntuSer tty1'. The user is prompted for a password and then greeted with the Ubuntu logo and system information. The terminal output shows the system load, usage of disk, memory, swap, and processes. It also displays the IP addresses for the network interfaces and the status of the expanded security maintenance for applications. The user is prompted to see additional updates and to enable ESM Apps to receive additional future security updates. The terminal output shows the user's login session and the prompt for the next command.

```
10.729468] cloud-init[1006]: | =00S.0 = . |
10.729492] cloud-init[1006]: | 0008.0 = + |
10.729559] cloud-init[1006]: | .0.+.++ = + |
10.729679] cloud-init[1006]: | .0. =. = |

10.730526] cloud-init[1006]: +----[SHA256]-----+

ubuntu 26.04.1 LTS ubuntuSer tty1
ubuntuSer login: adri
password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-38-generic x86_64)

* Documentation:  https://docs.ubuntu.com .0 |
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro

System information as of Fri Oct 2 04:42:33 PM UTC 2026

System load:          0.21
Usage of /:            14.2% of 24.44GB
Memory usage:          5%
Swap usage:            0%
Processes:             137
Users logged in:       0
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fe59:60b2

Expanded Security Maintenance for Applications is not enabled.

21 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

adri@ubuntuSer:~$
adri@ubuntuSer:~$ cd
adri@ubuntuSer:~$
adri@ubuntuSer:~$
```

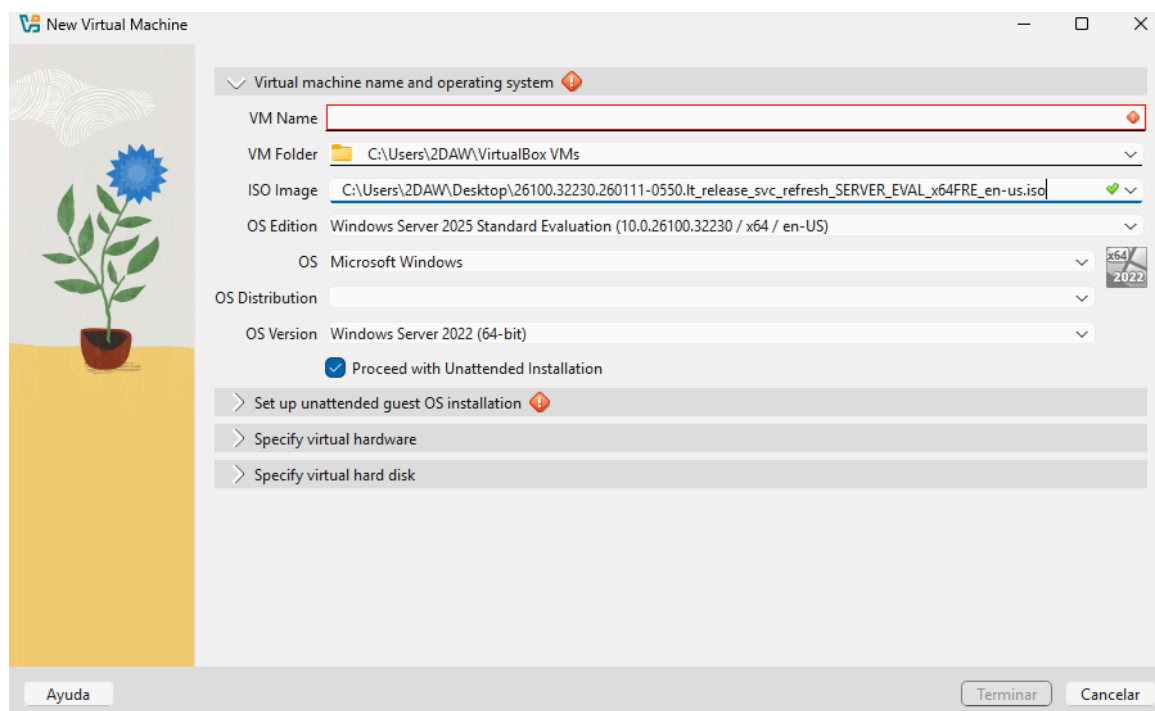
-INSTALACION WINDOWS SERVER

1.Buscamos la iso correspondiente.

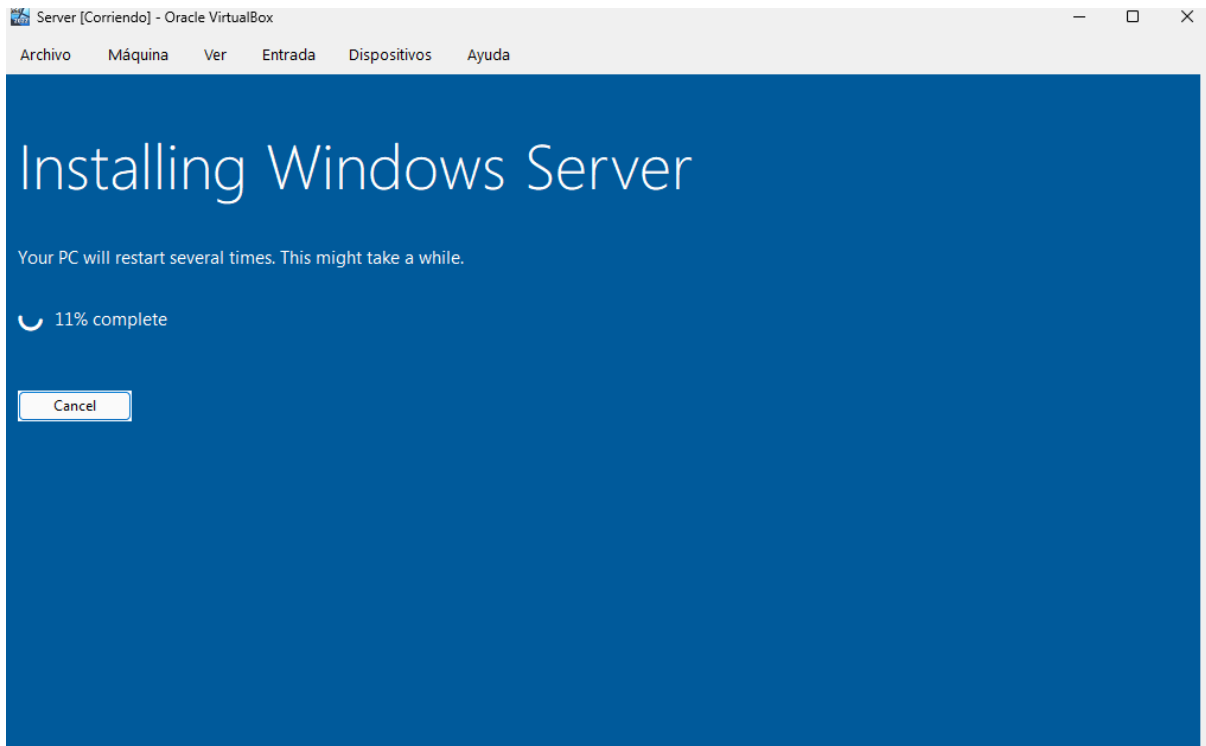
Selecciona tu descarga de Windows Server 2022.

Inglés (Estados Unidos)	Descargas ISO Edición de 64 bits >	Descarga de VHD Edición de 64 bits >	Prueba Azure Más información >	Crear una VM en Azure Más información >
Chino (simplificado)	Descargas ISO Edición de 64 bits >			
Francés	Descargas ISO Edición de 64 bits >			
Alemán	Descargas ISO Edición de 64 bits >			

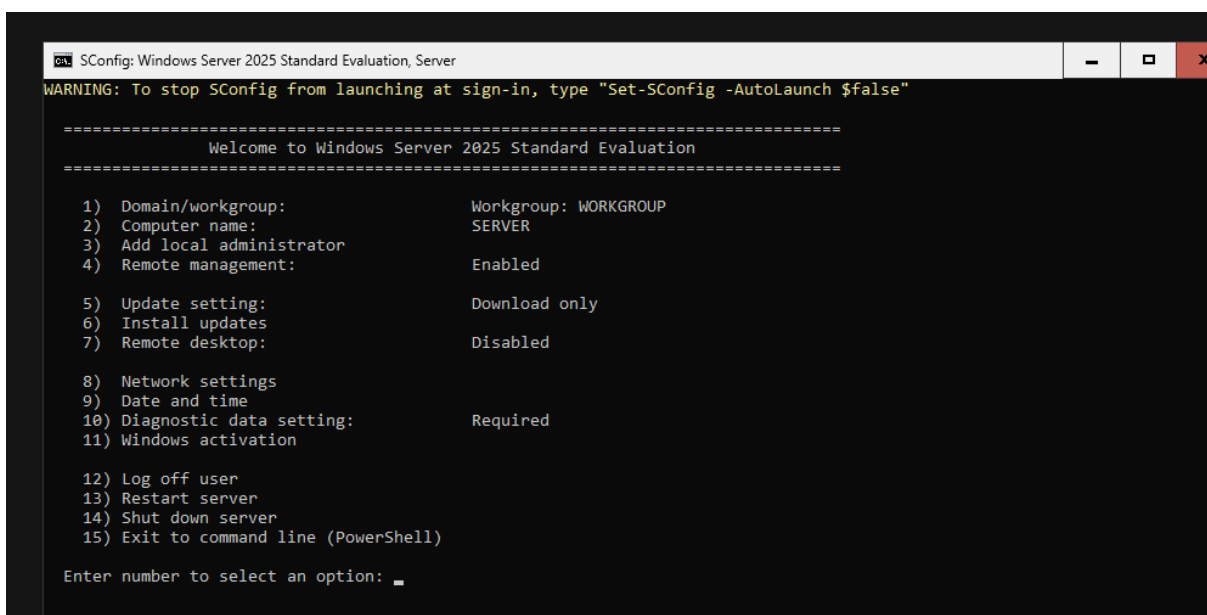
Agregamos la máquina nueva, esta vez con la iso de windows Server



Iniciamos la máquina virtual y empezará el proceso de descarga.



Elegiremos la opción 15 para salir del menú actual.



INSTALACION IIS EN WINDOWS SERVER.

Una vez en la terminal pondremos el siguiente comando

```
Administrator: C:\WINDOWS\system32\cmd.exe
WARNING: To launch Server Configuration tool again, run "SConfig"
PS C:\Users\vboxuser> Install-WindowsFeature -Name Web-Server -IncludeManagementTools

Collecting data...
10%
[oooooooooooo]
```

Verificar que el servicio IIS esté en ejecución

```
PS C:\Users\vboxuser> Get-Service w3svc

Status      Name      DisplayName
-----
Running     w3svc     World Wide Web Publishing Service

PS C:\Users\vboxuser>
```

Versión corregida (Este comando funciona si o si.

```
PS C:\Users\vboxuser> Invoke-WebRequest -Uri http://localhost -UseBasicParsing

StatusCode      : 200
StatusDescription : OK
Content         : <!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Strict//EN"
                  "http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd">
                  <html xmlns="http://www.w3.org/1999/xhtml">
                  <head>
                  <meta http-equiv="Content-Type" cont...
RawContent      : HTTP/1.1 200 OK
                  Accept-Ranges: bytes
                  Content-Length: 703
                  Content-Type: text/html
                  Date: Tue, 06 Oct 2026 14:50:33 GMT
```

7. Instala Docker en Ubuntu y despliega un servidor web Apache que escuche por el puerto 8081.

Primero hay que instalar el docker por lo que debemos de usar el siguiente comando.

```
ubuntu@Ubuntu:~$ sudo apt install docker.io -y
```

Una vez instalado comprobamos la versión del Docker.

```
ubuntu@Ubuntu:~$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu4.1
ubuntu@Ubuntu:~$
```

Comprobamos el estado en el que se encuentra el servicio con este comando.

```
ubuntu@Ubuntu:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-09-30 16:41:58 UTC; 2min 44s ago
     Invocation: 7c2e48fd5445451f8d3aab0fd9792f9e
   TriggeredBy: ● docker.socket
     Docs: https://docs.docker.com
    Main PID: 11252 (dockerd)
      Tasks: 9
     Memory: 40.4M (peak: 42.3M)
        CPU: 269ms
    CGroup: /system.slice/docker.service
            └─11252 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Sep 30 16:41:58 Ubuntu dockerd[11252]: time="2026-09-30T16:41:58.549154528Z" level=info msg="Rest
Sep 30 16:41:58 Ubuntu dockerd[11252]: time="2026-09-30T16:41:58.579633688Z" level=info msg="Dele
Sep 30 16:41:58 Ubuntu dockerd[11252]: time="2026-09-30T16:41:58.586511028Z" level=info msg="Dele
Sep 30 16:41:58 Ubuntu dockerd[11252]: time="2026-09-30T16:41:58.861245801Z" level=info msg="Load
Sep 30 16:41:58 Ubuntu dockerd[11252]: time="2026-09-30T16:41:58.875322092Z" level=info msg="Dock
Sep 30 16:41:58 Ubuntu dockerd[11252]: time="2026-09-30T16:41:58.875405236Z" level=info msg="Init
Sep 30 16:41:58 Ubuntu dockerd[11252]: time="2026-09-30T16:41:58.895989411Z" level=info msg="Comp
Sep 30 16:41:58 Ubuntu dockerd[11252]: time="2026-09-30T16:41:58.906546650Z" level=info msg="Daer
Sep 30 16:41:58 Ubuntu dockerd[11252]: time="2026-09-30T16:41:58.906606058Z" level=info msg="API
Sep 30 16:41:58 Ubuntu systemd[1]: Started docker.service - Docker Application Container Engine.

ubuntu@Ubuntu:~$
```

Comprobamos ahora que docker funciona correctamente haciendo que haga un Hello World.

```
ubuntu@Ubuntu:~$ sudo docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
d5e71e642bf5: Download complete
Digest: sha256:5e23090353324d887c48ad5e5c56d294eab81588df9605b07d1afe895f9cc8f8
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/

ubuntu@Ubuntu:~$
```

Ahora procedemos a descargar la imagen oficial de Apache

```
ubuntu@Ubuntu:~$ sudo docker pull httpd
Using default tag: latest
latest: Pulling from library/httpd
7a3bdd751d95: Pull complete
6b37362b3da7: Pull complete
6db0cf540580: Pull complete
4f4fb700ef54: Pull complete
2f4636dca226: Pull complete
1320c4426edf: Pull complete
fdff2e4882b0: Download complete
0b4153e35dab: Download complete
Digest: sha256:03f858efb82c25cb0f9962946615cdcdaf26927cb971722149b563197cfd0fdd
Status: Downloaded newer image for httpd:latest
docker.io/library/httpd:latest
ubuntu@Ubuntu:~$
```

Para comprobar que se ha descargado correctamente ponemos el siguiente comando para ver todas las imágenes.

```
ubuntu@Ubuntu:~$ sudo docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
hello-world:latest	5e2309035332	25.9kB	9.49kB	U
httpd:latest	03f858efb82c	177MB	47.6MB	

```
ubuntu@Ubuntu:~$
```

Ahora debemos de crear el contenedor Apache en el puerto 8081.

```
ubuntu@Ubuntu:~$ sudo docker run -d --name apache8081 -p 8081:80 httpd
9136a7be9bcee7d3afb3f5cb51ef61de5a7bbb89fa1bcacb3a98370535ace72a
ubuntu@Ubuntu:~$
```

Comprobamos que se ha realizado el comando correctamente.

```
ubuntu@Ubuntu:~$ sudo docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
9136a7be9bce	httpd	"httpd-foreground"	55 seconds ago	Up 54 seconds	0.0.0.0:8081->80/tcp, [::]:8081->80/tcp	apache8081

```
ubuntu@Ubuntu:~$
```

Ahora debemos de buscar en el navegador nuestra ip con el puerto correspondiente y ya estaría.

